Dataset Facts

Dataset	Open Air Chicago Day Aggregations
Team's source snapshot	117,892 rows; 46 columns (Oct. 14 notes)
EDA selection	15 columns; cleaned Oct. 14 row count not supplied

Motivation

Creators	CDPH and UIC School of Public Health
Purpose	Local $PM_{2.5}$/NO_2 monitoring beyond federal AQI coverage
Grant 1	NIST grant 60NANB23D206 (2023–2026)
Grant 2	RHP Risk Management / ComEd / Exelon (2025–2026)

Composition

Observation unit	One sensor's 24-hour mean for one complete UTC day
Time window	Sep. 1, 2025–Oct. 14, 2026 (team-reported)
Geographic reach	Chicago community areas and federal monitoring sites
Key pollutants	$PM_{2.5}$ and nitrogen dioxide (NO_2)
Known missingness	Black-carbon modules offline; invalid readings excluded entirely

Collection Process

Monitoring network	277 non-regulatory sensors (summer 2025)
Method	Clarity sensors; 00:00–23:59:59 UTC averages
Extra modules	Six sensors also measure black carbon and local weather
Other locations	Federal monitoring sites for comparison/calibration

Preprocessing

Publisher checks	Automated Clarity quality-control screening
Direct readings	Temperature, humidity and wind need no cloud correction
Calibration	$PM_{2.5}$/NO_2: `.raw` vs cloud-calibrated `.value`
EDA cleaning	15 columns kept; incomplete rows dropped in notebook

Uses and Distribution

Appropriate uses	Neighborhood trends, research and federal-data comparisons
Not suitable for	Regulatory findings or personal-exposure estimates
Access	Public Chicago Data Portal; CSV download and API
License / terms	No specific license identified; City disclaimer applies

Maintenance and Affected Communities

Stewards	CDPH / UIC SPH; City publishes
Updates	Daily; modification date shown on portal
Team snapshot	October 14, 2026; portal accessed Oct. 27, 2026
People / equity	No personal records; polluted-area residents may be affected

Limitations. Values are non-regulatory, reflect sensor locations rather than personal exposure, and cover only reported/quality-checked sensor-days. Raw and calibrated pollutant values should not be conflated.

Sources: City of Chicago, Open Air Chicago Day Aggregations; Data.gov metadata; CS 418 team notes and `eda.ipynb`. Counts are snapshot-specific.

Supporting documentation and source notes

Dataset and provenance. Open Air Chicago Day Aggregations is published by the City of Chicago. The project was developed by the Chicago Department of Public Health (CDPH) and the UIC School of Public Health (SPH), supported by an advisory board, neighborhood/grassroots organizations, and government partners. It provides more localized $PM_{2.5}$ and NO_2 information than the federal air-quality-index network alone. The publisher documents 277 non-regulatory network sensors and additional monitors at federal sites. Dr. Serap Erdal at UIC SPH received NIST and RHP Risk Management / ComEd / Exelon funding.

Collection and missing data. Each row is the mean for one sensor from 00:00 to 23:59:59 UTC. All network sensors monitor $PM_{2.5}$ and NO_2; six also have black-carbon/weather modules, although the black-carbon modules were offline when the data launched. Temperature, humidity and wind are direct sensor values. For $PM_{2.5}$ and NO_2, `.raw` is uncalibrated and `.value` is cloud-calibrated. Invalidated measurements are not in the public aggregate; separate high-resolution readings are publicly published, while invalidated data require a request.

Analytical limitations. The file captures reported, quality-checked sensor-days, not every place or personal exposure in Chicago. The observations cannot establish causation between income, emissions and health. These non-regulatory monitors should not be treated as legal compliance evidence or exact personal-exposure measures. Decisions based on these records could especially affect residents of heavily industrialized or polluted neighborhoods.

Access, terms, and reuse. CSV and API access are public on the Chicago Data Portal. Our group accessed the portal on October 27, 2026. No dataset-specific license was identified in the team's materials. The City's disclaimer requires derivative applications to display the notice below and honor any additional agency terms; do not assume unconditional rights to relicense processed data. The dataset is refreshed daily.

City of Chicago disclaimer provided by the project team:

"This site provides applications using data that has been modified for use from its original source, www.cityofchicago.org, the official website of the City of Chicago. The City of Chicago makes no claims as to the content, accuracy, timeliness, or completeness of any of the data provided at this site. The data provided at this site is subject to change at any time. It is understood that the data provided at this site is being used at one's own risk."

Sources and links

- City of Chicago - dataset and documentation
- Data.gov - metadata, funding, fields, and public distribution
- CDPH - Open Air Chicago program page
- City of Chicago - data disclaimer
- Original CSV export endpoint
- eda.ipynb (Our group's EDA notebook on GitHub. Selection, Missing Value Filtering, Additional Aggregation)

Unresolved reconciliation: Our notes say **117,892** rows / 46 columns from Sep. 1, 2025 onward. The EDA notebook instead uses a **117,320**-row extract, **115,655** rows after 15-column complete-case filtering, and an example reading dated July 26, 2025. Its **985 matched / 15 unmatched** join counts are from a **1,000-row sample**; the unmatched names span across three federal colocation sites, not missing Chicago community areas.